TESS planet candidate around TIC 410056850

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Summary

Signal found by the bls search of a public-data TESS pipeline (SPOC 120 s PDCSAP light curves, sectors 101, 102, 103, 104). Final tier: **candidate: likely planet**. Catalogue cross-match (TOIs, CTOIs, published planets, SPOC TCEs, TESS eclipsing binaries): **TOI 7850.01** (TOI). CNN score (calibrated): 0.916. TRICERATOPS (10 Monte Carlo runs, mean ± error of the mean): FPP $= 0.0709 \pm 0.0064$, NFPP $= 1.3e-07$.

Host star (TIC v8)

RA, Dec (deg)	320.87374, -71.23859
TESS magnitude	12.7
Radius ($R_\odot$)	1.34 ± 0.0611
$T_{\rm eff}$ (K)	6543 ± 128
Density ($\rho_\odot$)	0.561 ± 0.137
Distance (pc)	773

Transit fit

Limb-darkened model (quadratic law, Kipping q_1, q_2; GPU float64, 1024 annuli, 1024x4096 tables, 15x supersampling), emcee with 48 walkers, 12000 steps (1087 burn-in), run until the chain exceeded 50 autocorrelation times. Medians and 68% intervals.

Mid-transit time T_0 ($\mathrm{BJD_{TDB}}$)	$2461157.30666^{+0.00096}_{-0.00098}$
Period P (days)	$10.60528^{+0.00034}_{-0.00033}$
Radius ratio $R_p/R_\star$	$0.0848^{+0.0020}_{-0.0028}$
Impact parameter b	$0.56^{+0.11}_{-0.24}$
Stellar density $\rho_\star$ (g cm^{-3})	$0.50^{+0.22}_{-0.12}$
Limb darkening q_1	$0.24^{+0.29}_{-0.14}$
Limb darkening q_2	$0.25^{+0.35}_{-0.18}$
Depth (ppm)	7798^{+195}_{-197}
Duration T_{14} (hours)	$5.25^{+0.11}_{-0.11}$
$a/R_\star$	$14.4^{+1.8}_{-1.3}$
Inclination (deg)	$87.8^{+1.1}_{-0.7}$
Planet radius ($R_\oplus$)	$12.36^{+0.67}_{-0.65}$
Semi-major axis (au)	$0.090^{+0.011}_{-0.009}$
Equilibrium temperature (K, $A = 0$)	1217^{+66}_{-69}
Insolation ($S_\oplus$)	366^{+86}_{-76}

Vetting

Test	Value	Pass condition	Result
Transit SNR (correlated noise)	43.1	≥ 7.1	pass
Transits observed	8	≥ 2 (periodic)	pass
Odd/even depth difference (σ)	0.641	≤ 4	pass
Secondary eclipse (σ)	2.49	≤ 3.8	pass
Ingress fraction (V shape)	0.147	≤ 0.4	pass
Planet radius ($R_\oplus$)	12.7	≤ 25	pass
Centroid source offset (arcsec)	0.717	4σ and $>21''$	pass
Difference image offset (arcsec)	1.3	4σ and $>10.5''$	pass
Difference image significance (σ)	0.428	–	–
CCD common mode (σ)	-0.968	≤ 4	pass
Crowding (CROWDSAP)	0.921	≥ 0.8	pass
Stellar density ratio (max/TIC)	1.02	≥ 0.2	pass

Caveats

- This is a candidate with statistical evidence, not a confirmed planet: confirmation needs high-resolution imaging, reconnaissance spectroscopy and radial velocities.
- TRICERATOPS was run without a high-resolution imaging contrast curve, so close bound companions are not excluded.
- The MCMC converged (chain longer than 50 autocorrelation times, stable estimate) (acceptance 0.17, 12000 steps, about 2402 independent samples); errors are scaled by the time-averaging factor $\beta = 1.02$ for correlated noise.
- Stellar radius and effective temperature come from TIC v8; if the TIC gives no error, 7% (radius) and 150 K were assumed.

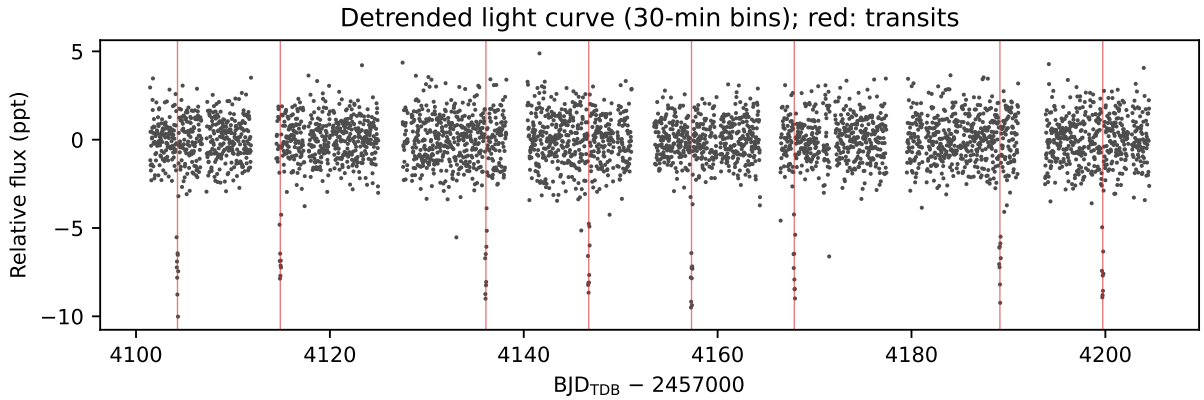


Figure 1: Detrended SPOC PDCSAP light curve (transit-protected biweight).

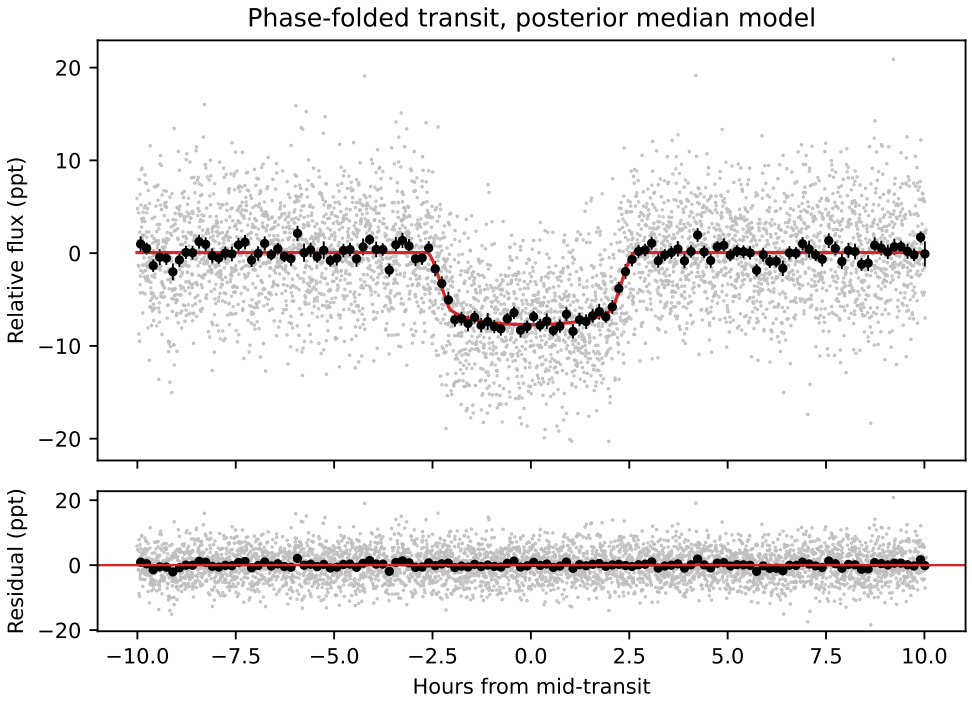


Figure 2: Phase-folded transit with the posterior median limb-darkened model.

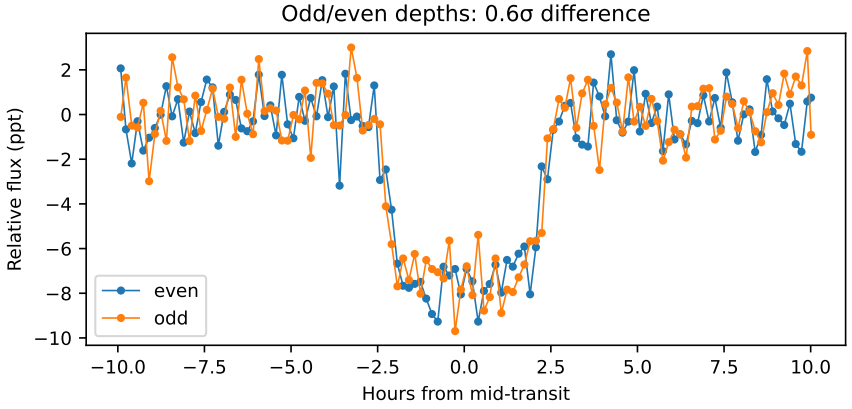


Figure 3: Odd and even transits, binned to 10 minutes.

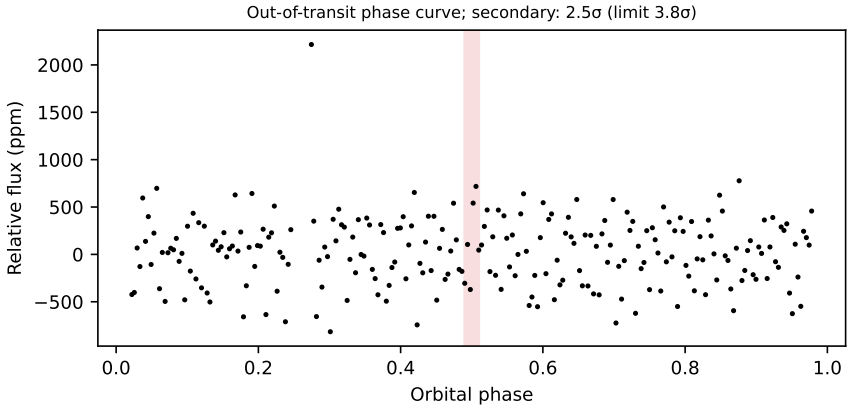


Figure 4: Phase curve; the shaded band is the secondary eclipse window for a circular orbit.

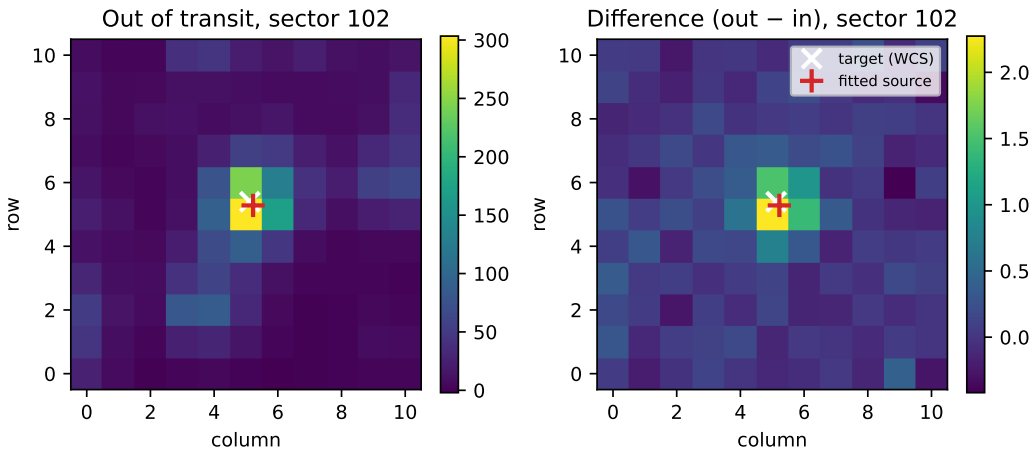


Figure 5: SPOC target pixel file: out-of-transit image and difference image. White cross: target catalogue position; red plus: fitted source.

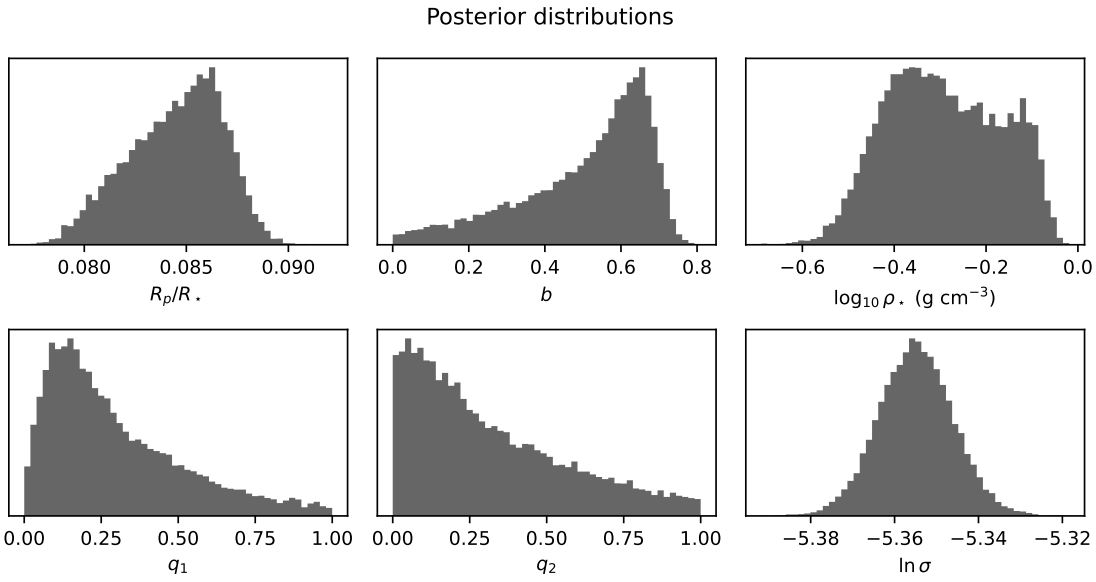


Figure 6: Posterior distributions of the fitted parameters.